ENGINEERING DESIGN AND MODELLING

NAME: DHARSHAN.V

EX NO : 1.1 , 2D Sketch Design using FUSION 360.

1. AIM

To create a 2D sketch design using Autodesk Fusion 360 by applying sketch tools, circles, arcs, lines, trimming, mirroring, dimensions, and constraints according to the given engineering drawing.

2. SOFTWARE AND TOOLS REQUIRED

1.Software: Autodesk Fusion 360

2.Tools required:

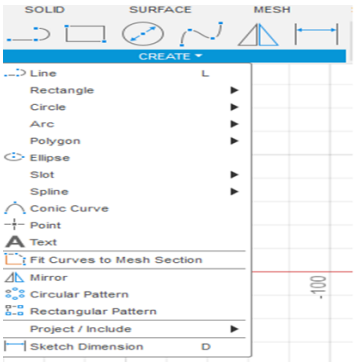
- Create Sketch
- Circle Tool
- Line Tool
- Arc Tool
- Sketch Dimension
- Trim Tool
- Mirror Tool
- Constraint Tools

3. PROCEDURE

Step 1: Open Autodesk Fusion 360 on the computer.

Step 2: Create a Sketch.

The sketch workspace will open with the following features.



Step 3: Draw the Centreline.

Select the Line tool.

Draw a vertical construction line through the origin.

Use this line as the centreline for the design.

The centreline helps maintain symmetry.

Step 4: Draw the Top Circle.

Select the Circle tool.

Choose Centre Diameter Circle.

Place the centre on the vertical centreline.
Draw the outer circle with a diameter of 32 mm.
Draw another concentric circle with a diameter of 20 mm.
Use Sketch Dimension (D) to enter the exact diameters.

Step 5: Draw the Main Body.

Use the Line and Arc tools to create the main body.
Connect the top circular portion to the middle section.
Draw the curved sides according to the given drawing.
Apply the specified radius of R64 mm.
Use Sketch Dimension to enter the required dimensions.

Step 6: Create the Side Portions.

Draw the curved portions on the left and right sides.
Use the Arc tool to create the required curves.
Apply the radii R20 mm, R10 mm and R7 mm where indicated.
Maintain symmetry about the vertical centreline.

Step 7: Draw the Side Holes.

Select the Circle tool.
Draw the circular outer portions at both sides.
Draw the inner holes according to the given drawing.
Position the hole centres 47 mm from the vertical centreline on each side.
Apply the required dimensions.

Step 8: Draw the Bottom Circle.

Select the Circle tool.
Place the centre on the vertical centreline.
Draw the outer circle with a diameter of 20 mm.
Draw the inner circle with a diameter of 10 mm.
Align both circles concentrically.

Step 9: Draw the Central Opening.

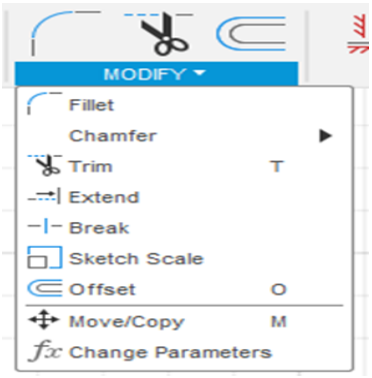
Use the Line and Arc tools to draw the shaped central opening.
Create the six-lobed internal profile as shown in the reference drawing.
Apply the required dimensions and the R6 mm radius.
Ensure that the opening is positioned at the centre of the design.

Step 10: Use the Mirror Tool.

Select the geometry on one side of the vertical centreline.
Open the Sketch menu.
Select Mirror.
Select the geometry to mirror.
Choose the vertical construction line as the mirror line.
Click OK to create the symmetrical portion.
If both sides have already been drawn, do not mirror them again.

Step 11: Use the Trim Tool.

Select the Trim tool or press the T key.
Click on unwanted portions of lines and arcs.
Remove overlapping segments carefully.
Ensure that the required outline remains intact.
Trim and Mirror using the given modifications.



Step 12: Apply Dimensions and Constraints.

Press D to activate Sketch Dimension.

Enter the required diameters, radii and distances.

Apply Coincident constraints to connect geometry.

Apply Tangent constraints where required.

Check the symmetry of the design.

Step 13: Check the Sketch.

Inspect the complete drawing.

Check whether all dimensions are correct.

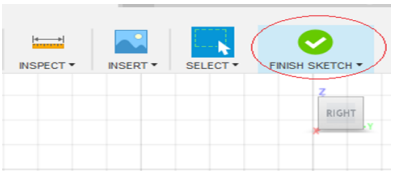
Ensure that there are no unwanted lines or overlapping arcs.

Check that the required closed profiles are properly connected.

Step 14: Finish the Sketch.

Click Finish Sketch.

Save the design.



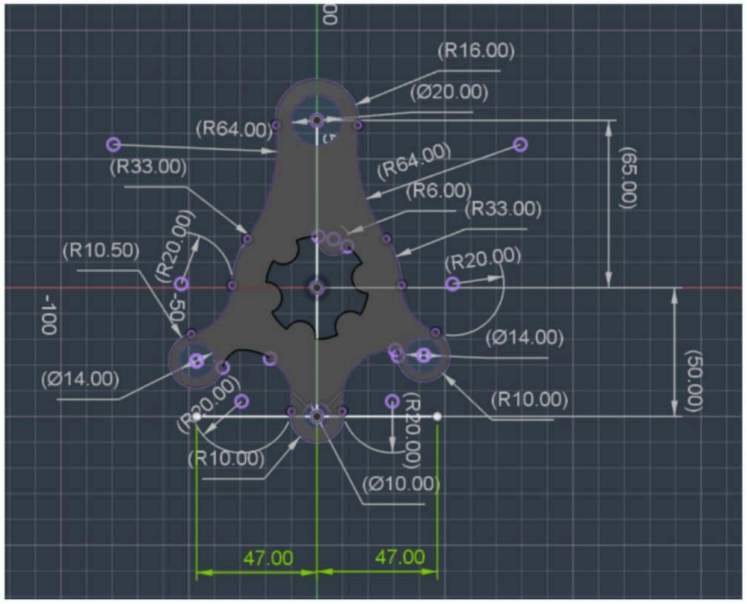
Step 15:Extrude the saved design.

Using Extrude select the sketch and extrude it.

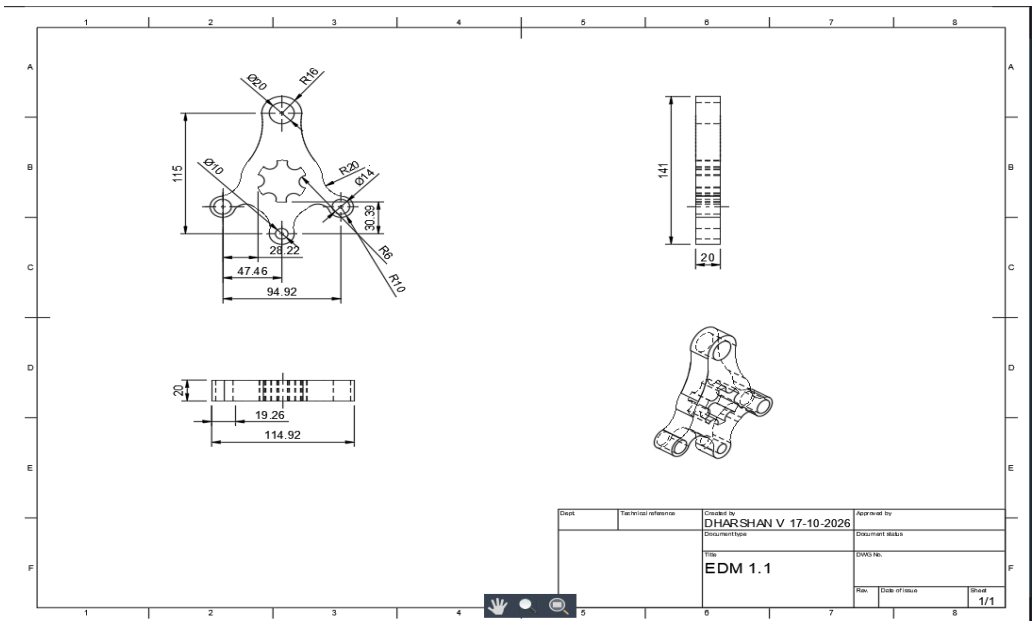
Now,make a sheet design for the 2D model.

4.OUTPUT

1.Sketch.



2.Design Sheet .



5. RESULT

Thus,the given model is designed and drafted using the Autodesk Fusion 360 using sketch tools, circles, arcs, dimensions, constraints, trimming and mirroring according to the given engineering drawing.

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